

Circuits & Systems — Study Pack (MTE)

ECE2107 · B.Tech ECE Sem 3 · follows the professor's MTE syllabus (dated 10/9/26) · built from two sets of class notes · every answer checked by circuit simulation

Priority order: if time is short, study in this order

Ranked by how much the professor stressed each topic: the number of board questions worked in class, and how many notes pages it took. There is **no past paper yet**, so this is an estimate. The one hard fact is the syllabus, and every item on it can come up.

Rank	Study	Why it ranks here	Do first	Rough time
1	06 Thévenin & Norton	The professor's longest worked question (dependent source, nodal route); it appears in both notes sets	06 §2–5, then mock B10	60 min
2	05 Superposition	2 board questions + the "limitation" question the professor highlighted	05 §1–5, mock B7–B8	40 min
3	08 First order	4 board questions (switching at $0^-/0^+/\infty$, τ , RC and RL); Laplace and test signals are separate syllabus items	08 §1–4, then §6–7	60 min
4	07 Maximum power transfer	2 board question sets, one of them six parts long; the derivation is a likely theory question	07 §2–3, mock B12–B14	30 min
5	04 Nodal & mesh	A syllabus item, and 06 depends on it (the professor solved V_{Th} by nodal)	04 §1–4	40 min
6	09 Second-order RLC	A syllabus item with four named cases; no board question yet	09 §3–4, exam form	40 min
7	02 Dividers, power & energy	4 short board questions, all quick marks	02 §1–4	25 min
8	03 KVL, KCL, source transformation	Short; used inside every other method	03 §1–4	20 min
9	01 Elements & sources	Theory only (classification, dependent sources, lumped vs distributed)	01 exam form	20 min
AC prereq	04b AC in 15 minutes	Phasors, j and angles. The AC parts of 05, 06, 07 and mock B11 all use it	read it before any AC section	15 min
last	10 Mock paper	Closed-book, after the rest	all	2–3 h

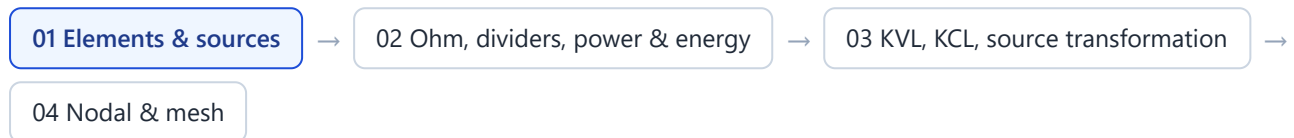
Two rules for using the ranking:

- **Short on time:** go down the list and do only the "Do first" column, then the matching mock-paper items.
- **AC ($\angle$, j) is new to you:** read 04b once (15 min) before any section marked AC.
- **Stuck on 06:** it builds on 05 (switching sources off) and 04 (nodal). Go back one step and don't skip ahead.

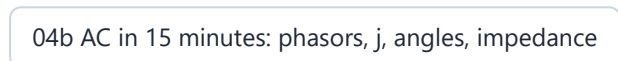
The path through this pack

The order is the professor's MTE syllabus, item by item.

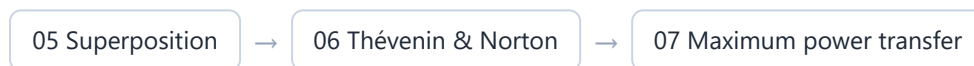
Part I — elements, laws, analysis



Bridge — AC basics (before Part II's AC sections)



Part II — theorems (DC and AC)



Part III — transient analysis



Each file needs the ones before it. 04 is KCL/KVL (03) written as a method. 06 uses 05's "switch a source off" rule, 04's nodal analysis and 01's dependent sources. 07 is built on 06. 08 uses 06 for the time constant; 09 uses 08's Laplace route.

File	Syllabus item	What it teaches	Board Qs
01	1 · elements (R, L, C, IDS, DS)	active vs passive, independent and dependent sources (with devices), R, L, C laws, what can't jump, lumped vs distributed	—
02	2 · laws	Ohm, voltage and current division (DC and AC), power sign convention, energy, ladders	4
03	3 · KVL, KCL, source transformation	the two laws, power balance, source transformation	1 + 1 textbook
04	4 · nodal and mesh	both methods, with current sources and dependent sources	1 + 1 textbook
04b	tool for the AC parts of 5	AC source as a cosine, phasors, j , rectangular $\leftrightarrow$ polar, impedance of R, L, C, the 4-step recipe	textbook
05	5 · superposition	method, the cases where it fails, AC sources at different frequencies	2 + 1 textbook
06	5 · Thévenin & Norton	with a dependent source; with AC impedances	1 + 1 textbook
07	5 · maximum power transfer	$R_L = R_{Th}$, efficiency, conjugate match (worked)	2 sets + 1 textbook
08	6 · first order	$0^- / 0^+ / \infty$, τ , the one formula, Laplace, test signals δ , u , r	4 + 2 textbook
09	7 · second order	series RLC: over-, critically, under-damped, undamped	textbook
10	all	one paper: 12 theory prompts + 20 numericals	14 + 6 textbook

Where the questions come from

A question that opens a section is **quoted from the professor's board** where one exists, and tagged with its notes page: **Class p.7** = first notes set (the 23-page scan), **Set 2 p.9** = second notes set (22 photos). Where the syllabus names a topic the class has not yet worked a question on, the pack uses a standard textbook-style problem, tagged **textbook**.

	Base	Status
P1	The professor's board questions (notes sets 1 and 2)	14 questions , all used
P2	Textbook problems	7 , written for syllabus items with no board question (KVL, nodal/mesh, AC theorems, Laplace, test signals, RLC). Values chosen so answers are clean; all checked by simulation

Marks per question: **not known** (no hand-out, no past paper yet).

How each file runs

Part	What happens
Map	the file's pieces, in order
Build	each step opens with a real question → you guess → the answer teaches the idea → a Check
Exam form	what to reproduce in the exam: statements, procedures, the formula sheet
Traps	the mistakes that cost marks
Self-test, Answers	answers are on the last page(s) of every file

THE ONE RULE THAT MAKES THIS WORK

When you reach a **Guess first** box, write a guess on paper **before** reading on: one line, however wrong. A wrong guess that gets corrected sticks better than a right answer you only read.

Mark every Check and Self-test item:

- **clean** → revisit in 3 days
- **with struggle** → revisit tomorrow
- **wrong** → redo today, after the file

Units and symbols used everywhere

- Uppercase **V, I** = DC (constant) values. Lowercase **v, i** = values that change with time. Complex values such as $5 - j5$ V are **phasors**: amplitude and phase in one number.
- Resistances in **kΩ** with currents in **mA** give voltages in **V** directly: $2 \text{ k}\Omega \times 3 \text{ mA} = 6 \text{ V}$. Most questions use this.
- **R_{Th}, V_{Th}** = Thévenin resistance and voltage. **I_N** = Norton current. **R_L** = load resistance.
- **0⁻** = just before the switch moves. **0⁺** = just after. **∞** = long after (steady state).

Where the two notes sets disagree with the pack

Topic	Notes say	Pack says	Why
Superposition, two 5 V sources (05 §5)	set 2: $0 + 0 = 0$ A, "LIMITATION"; set 1: 10 mA	0 A by superposition, true 5 mA	set 2 is the correct use of the rule and shows why it fails; the pack follows set 2
Thévenin with 2I (06 §2)	set 2: $V_{Th} = 5.0025$ V	20/3 V	set 2 reads 2I as $2 \Omega \times I$; the pack reads $2 \text{ k}\Omega \times I$. Both are right for their reading. Ask the professor
RL switch that opens (08 §5)	$I_R(0^+) = 0$, $I_L(0^+) = 10$ mA	cannot both hold in one series loop	ask the professor which path the current takes

How this pack was checked

- Facts: checked against the textbook-sourced circuits knowledge base (Hayt, Van Valkenburg, Sudhakar & Shyammohan) and Alexander & Sadiku.
- Numbers: every circuit in this pack was solved a second time by computer (nodal analysis and time simulation), and the answers here match. The script is `src/verify_mte.py`.
- Where the class notes had a slip of the pen, the pack uses the corrected version and says so in a Note.